

INTRODUCTION TO DATA SCIENCE AND PROGRAMMING

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DATA SCIENCE



WHAT IS DATA
SCIENCE?

WHAT IS DATA SCIENCE?

- What subjects do students enjoy the most?
- How do students prefer to learn?
- Do students feel encouraged to ask questions during the lesson?
- How well are students performing on assessments?
- What feedback do students have about lessons?

DATA SCIENCE

- Data science is a **discipline** of science
- It focuses on the process of using data to **understand a topic** and **make decisions**
- It involves
 - Gathering data – **data collection**
 - Displaying data – **data visualisation**

```
001111110000001111100100110001101110010100111111111101
1110111111000011101011111101100011110111111010101100110
1101111100001110001101010010111010011111000110001000110
1100011101111011111110111111111010001111110111011111110
11111001101110111110001111011101011101000111111101111110
110011011111101100010010000001011101110111011111111111
11111000111111000111111111111110011110011111100101111111
0111111011001110111110111111101011111110111111101100111010
1111111110001111111001010010100011111011110110100111101
0001111111111011000101000011110010000000111100100100011
10111110011111111101010111111000101110111000001001111110
0111000011110111011111100111111111111001100000110111
1011101101000010011001100011101110000110010000001100111
1101100110001010111101101000111110100011010111110010111
1111111110011000111100111101010000110100111000100100010
1110100001011100111111100000111110111111011110101111100
1110010011010111111111110011111111111111111111001010111111
0000000001111111100001100111101100100011011011010110001
010001001111111100111111111111110011111000111110011110101
010001111001001111111100001011110111001101111110110011111
```

PROGRAMMING

PROGRAMMING

- A **programming language** is a special **set of rules** and instructions that allow humans to **communicate with computers**
- We can give them **commands** to perform specific tasks
- Using programming languages, we can complete tasks at lightning-fast speeds
- A **program** refers to code that does a task

PYTHON

- The programming language that we will be using is Python
- Why is Python a good language to learn?
 1. It is the most popular language in the world
 2. It plays an important part in **AI**
 3. It is **easy to understand**

```
print("Hello, world!")
```

```
.global _start
.align 2

_start:
    b _printf
    b _terminate

_printf:
    mov X0, #1
    adr X1, helloworld
    mov X2, #12
    mov X16, #4
    svc 0

_reboot:
    mov X0, #1
    mov X16, #55
    svc 0

_terminate:
    mov X0, #0
    mov X16, #1
    svc 0

helloworld: .ascii "hello world\n"
```

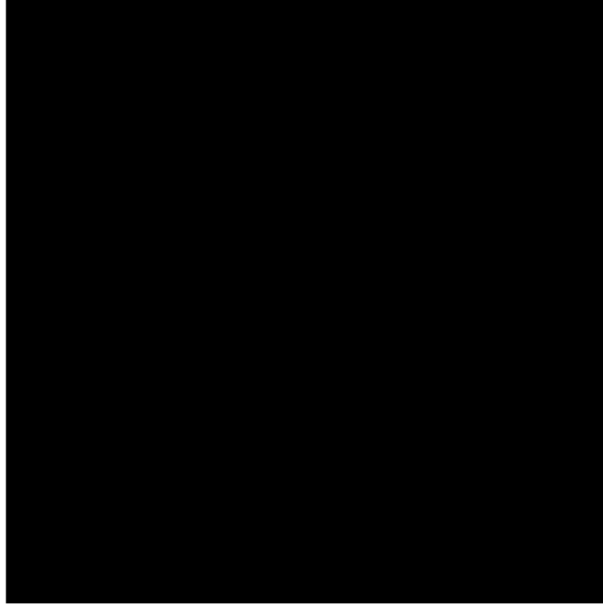
PYTHON

INTRODUCTION TO FUNCTIONS

- To help us solve problems, we will be making use of two functions
 - `print()` – this lets us give information **out from a computer**
 - `input()` – this lets us take information **in from a user**
- `print()` and `input()` are called **functions**
- Functions refers to **code** that performs a specific task

`print()`

Takes "Hello, world!" in



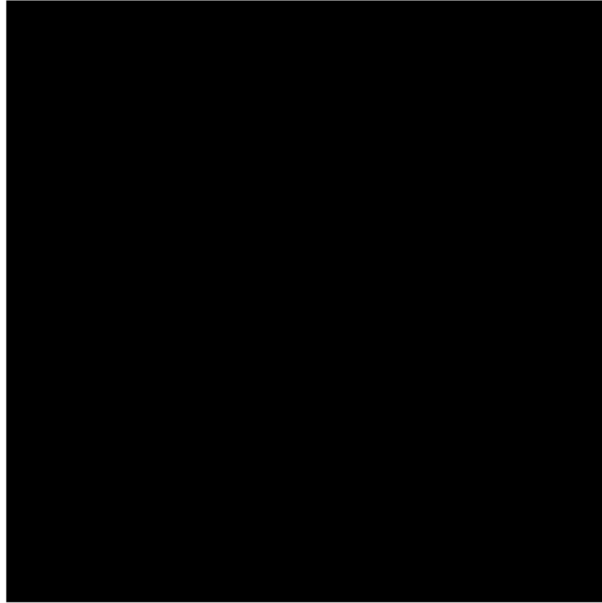
prints out "Hello, world!"



INTRODUCTION
TO FUNCTIONS

input()

Asks "What is your age?"



stores the input as a
string



INPUT()

INTRODUCTION TO FUNCTIONS

- A program can contain **functions** – each function doing a specific task that helps to complete the overall goal of the program
- For example, a program that records student information could have:
 - A function to record names
 - A function to record grades

ASSIGNMENT

WHAT IS ASSIGNMENT?

- **Assignment** is when give a value to a variable – you are **assigning** it a value
- A **variable** in programming is a **container** used to store **values** that can changed or be used later in a program
- They are called variables because you can **change**, or vary, what they store

VARIABLES

- You can name a variable **anything**
- When you assign a value to a variable, the variable must be on the **left**
- We write the names of variables using **lowerCamelCase**, to help us read the variable's name

```
myAge = 20
myName = "Liam"
gameOver = False
price = 19.99

exampleofvariablenamesinpython = "name"
examplesOfVariableNamesInPython = "name"
```

VARIABLES

TYPES

- **Types** are **labels** that tell the computer **what kind of data** we are giving it
- Every language uses types, but different programming languages might have different types

TYPES

- An **int** is short for integer – an integer is **whole positive or negative number** e.g. -113, -8, 35, 100000000
- A **String** is a sequence of characters e.g. “Hello, world!”, “my name is Liam”
 - It must be surrounded by **quotation marks**, “”
- A **boolean** can either be **true** or **false**
- A **float** is a number with a **decimal point** e.g. 123.45, -4.12, 3.1415

```
# A comment is any code that is ignored by the computer  
# Anything written after this hashtag is a comment
```

```
myAge = 20  
    # Type: int
```

```
myName = "Liam"  
    # Type: String
```

```
gameOver = False  
    # Type: boolean
```

```
price = 19.99  
    # Type: float
```

TYPES

TYPES

- You can use the **type()** function to check the type of a variable
- You can convert data from one type to another by **casting** it – wrap that variable in the type you want

```
# type() function

myName = "Liam"
myGrade = 73
pi = 3.1415
amIDone = False

print(type(myName))
print(type(myGrade))
print(type(pi))
print(type(amIDone))

# Casting

myAge = int(input("What is your age? ")) # Example of casting
birthYear = 2024 - myAge
print("You were born in " + str(birthYear)) # Example of casting
```

TYPES

Arithmetic Operators

Operator	Meaning	Example
+	Addition	$4 + 7 \longrightarrow 11$
-	Subtraction	$12 - 5 \longrightarrow 7$
*	Multiplication	$6 * 6 \longrightarrow 36$
/	Division	$30 / 5 \longrightarrow 6$
%	Modulus	$10 \% 4 \longrightarrow 2$
//	Quotient	$18 // 5 \longrightarrow 3$
**	Exponent	$3 ** 5 \longrightarrow 243$

ARITHMETIC
OPERATORS

ARITHMETIC OPERATORS

- The **modulus** operator (%) returns the remainder of a division between two numbers
- The **quotient** operator (//) gives the number of times one number can be divided by another without considering the remainder
- The **exponent** operator (**) raises a number, called the **base**, to the power of another number, called the **exponent**

PRACTICE